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Estimation of N_2 fixation by isolates of *Azotobacter* isolated from agriculture field of Madhepura district

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Abstract- *Azotobacter* sps. are able to fix atmospheric nitrogen. They are considered as plant growth promoting rhizobacteria (PGPR). In the present study, seven isolates of *Azotobacter* were isolated from rhizospheric soil collected from wheat and maize field. N_2 fixation capacity of all isolates was tested at different temperature and pH. Out of 7 isolates 4 belongs to *A. chroococcum* and 3 belongs to *A. vinelandii*. All isolates were able to grow in nitrogen free medium and also able to fix nitrogen. Nitrogen fixation capacity of all isolates was maximum at a temperature of 34°C and alkaline pH. Isolate ACW3 of *A. chroococcum* produced 11µg/g soil at 34°C temperature and pH-8 which was maximum among all isolates. At temperature below 34°C or above it decreased nitrogen production. Similarly at low pH, Nitrogen production decreased.

Keywords: ***Azotobacter, A. chroococcum, A. vinelandii,*** **Nitrogen fixation, pH, temperature.**

INTRODUCTION

Biological nitrogen fixation is an important biological process which is the reduction of atmospheric nitrogen in two mol. of ammonia. Nitrogen fixation is exclusively mediated by a large number of symbiotic and non-symbiotic bacteria as well as by Cyanobacteria. Among non-symbiotic bacteria *Azotobacter, Azospirillum, Pseudomonas*, etc. are most common in agriculture field. *Azotobacter* is capable of fixing 20kgN/h per year. Species of *Azotobacter* are found in soil and rhizosphere of many plants. Several species of *Azotobacter* have been described from different area which include *A. chroococcum, A. vinelandii, A. paspali, A. agilis, A. macrocytogens* and *A. beijerinckii.*[1] Beside nitrogen fixation *Azotobacter* sps. are capable of producing IAA, Gibberlins and Vitamin B which benefits plants in their growth.[2] *A. chroococcum* is widely spread in agriculture soil of temperate area. They may be easily discovered from rhizosphere of cereals.[3] In addition to nitrogen fixation, IAA production and vitamin B production *Azotobacter* is able to decompose many pollutants. Wang *et al.* (2009)[4] reported that *Azotobacter* is able to decompose a wide range of chlorinated phenols. Eklund *et al.* (2017)[5] reported that *A. chroococcum* in rhizosphere of cucumber and tomato increased growth and seedling germination.

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MATERIAL & METHODS

Soil samples were collected from rhizosphere of maize and wheat crops. Samples were labelled with name of site, crop and date of collection. Samples were brought to the laboratory.

Isolation of Bacteria:

One gram soil sample was dissolved in 10ml distilled water and diluted up to 10^{-5} dilution. Diluted sample was inoculated in Ashby's medium and incubated at 34°C for

24 hours. Single colony from culture was transferred in fresh medium to get pure culture.

Identification of Bacteria:

Bacteria were identified on the basis of their morphology, Gram staining, Biochemical tests and Sugar fermentation tests as described in Bergey's manual of Systematic Bacteriology.

Bio-Chemical Test:

Catalase test: For Catalase test, one loopful culture was taken on a sterile plain slide and one drop of 3% H_2O_2 was added on it. It was mixed thoroughly. Appearance of bubble or froth indicate positive result. If not any bubble or froth appears, the result is negative.

Indole test: Some bacteria have Enzyme Tryptohanase which break down Tryptophane to produce Indole. Production of Indole is tested using Kovac's reagent.

Bacterial inoculum was taken from culture and inoculated in Peptone water in a test tube. Culture was incubated at 37°C for 48 hours. 0.5ml Kovac's reagent was added and shaken gently. A red color ring shows positive result.

Oxidase test: Two drops of reagent are dropped in agar plate. The appearance of blue color shows positive result. The reagent for oxidase test is prepared as follow: 1%N, N, N', N'-tetramethyl-p-phenylenediamine dihydrochloride is dissolved in distilled water.

Sugar fermentation test was performed for the fermentation of glucose, fructose, sucrose, lactose, mannitol, raffinose and arabinose. For sugar fermentation test, 1% sugar was supplemented in peptone broth and the phenol red was added as indicator. All isolates were separately inoculated in test tube containing medium and incubated for 3-4 days. Color change was observed. If the color changes from pink to yellow, the result was considered as positive otherwise negative.

Estimation of Nitrogen: All isolates were inoculated in Ashby's broth. One set of broth was kept without inoculation. Culture was incubated for 10 days at 37°C temperature. Percentage Nitrogen was determined by Kjeldahl method.

After 10 days incubation, 250ml isolate containing broth was transferred in Kjeldahl microflask. Digestion salt mixture and 3ml Conc. Sulphuric acid was added and digested on sand bath. After digestion, 10ml distilled water was added and cooled (Digestion salt mixture contain Potassium sulphate, Copper sulphate and metallic selenium in a ratio of 50:10:01).

Digested sample was added in Kjeldahl distillation apparatus. 10ml 40% NaOH was added in distillation flask. 10ml 40% Boric acid and 3 to 4 drops Bromo-cresol reagent was taken in a conical flask. The conical flask was connected to the condenser of Kjeldahl apparatus. After distillation, the content of conical flask was titrated against HCl. Appearance of Pink color showed end point. Similar experiment was conducted with Ashby's broth without isolate which was the blank. Percentage Nitrogen was estimated by the formula:

$$\% \, Nitrogen = \frac{Sample\ titre - Blank\ titre}{Volume\ of\ sample} X\ Normality\ of\ HCl\ X\ 14X100$$

RESULT

Altogether seven isolates were identified, of which four belong to *A. chroococcum* and three belong to *A. vinelandii*. List of isolates is mentioned in Table 1.

Table 1- List of isolates

Soil code	Crop	Isolate	Sps.
KSW	Wheat	ACW1	*A. chroococcum*
KSW	Wheat	AVW1	*A. vinelandii*
CHW	Wheat	ACW2	*A. chroococcum*
BHM	Maize	ACW3	*A. chroococcum*
BHM	Maize	AVW2	*A. vinelandii*
JEW	Wheat	ACW4	*A. chroococcum*
SHM	Maize	AVW3	*A. vinelandii*

Table 2- Morphology and gram staining of isolates

Isolate	Cell morp.	Colony morp.	Gram staining
ACW1	Oval	Circular, white, elevated	-ve
ACW2	Oval	Circular, white, elevated	-ve
ACW3	Oval	Circular, white, elevated	-ve
ACW4	Oval	Circular, white, elevated	-ve
AVW1	Spherical	Circular, white, elevated	-ve
AVW2	Spherical	Circular, white, elevated	-ve
AVW3	Spherical	Circular, white, elevated	-ve

Table 3- Biochemical test of isolates

Isolate	Cat	Oxi	Citra	Indole	MR	VP
ACW1	+ve	+ve	+ve	-ve	-ve	-
ACW2	+ve	+ve	+ve	-ve	-ve	-
ACW3	+ve	+ve	+ve	-ve	-ve	-
ACW4	+ve	+ve	+ve	-ve	-ve	-
AVW1	+ve	+ve	+ve	-ve	-ve	-
AVW2	+ve	+ve	+ve	-ve	-ve	-
AVW3	+ve	+ve	+ve	-ve	-ve	-

Table 4- Sugar fermentation test of isolates

Isolate	Glu	Lac	Fruc	Sucr	Manni	Arabi
ACW1	+ve	-ve	-	+ve	+ve	-ve
ACW2	+ve	-ve	-	+ve	+ve	-ve
ACW3	+ve	-ve	-	+ve	+ve	-ve
ACW4	+ve	-ve	-	+ve	+ve	-ve
AVW1	+ve	-ve	-	+ve	+ve	-ve
AVW2	+ve	-ve	-	+ve	+ve	-ve
AVW3	+ve	-ve	-	+ve	+ve	-ve

All isolates were able to grow in nitrogen free Ashby's medium. Isolates of *A. chroococcum* were oval in shape while isolates of *A. vinelandii* were spherical. Colony morphology of all isolates were observed as cirucular, white and elevated. All isolates were gram negative (Table 2), catalase, oxidase and citrate positive, indole and MR negative (Table 3). Sugar fermentation test for all isolates showed that all isolates were glucose, sucrose and mannitol positive while lactose and arabinose negative (Table 4).

N_2 production by different isolate was examined at temperature of 20^{0}C, 34^{0}C, 40^{0}C and at pH 6, 7, 8 and 9. Maximum N_2 production was observed at pH-8 and pH-9 when temperature was 34^{0}C. At low and high temperature N_2 production decreased. At a temperature of 34^{0}C and alkaline pH, maximum N_2 production was observed by isolate no. ACW3 collected from maize field of Bhelwa and minimum N_2 production was observed by isolate no. AVW3 collected from maize field of Shankarpur. Effect of temperature on N_2 production is shown in Fig. 1 and effect of pH on N_2 production is shown in Fig. 2.

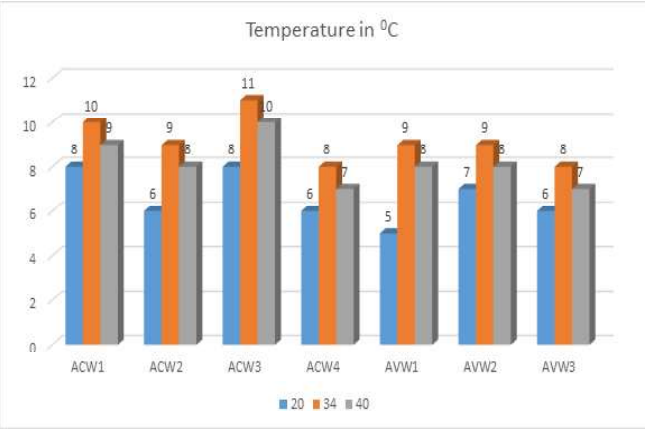


Fig. 1- Graph showing estimation of N_2 production in µg/g soil at different temperature.

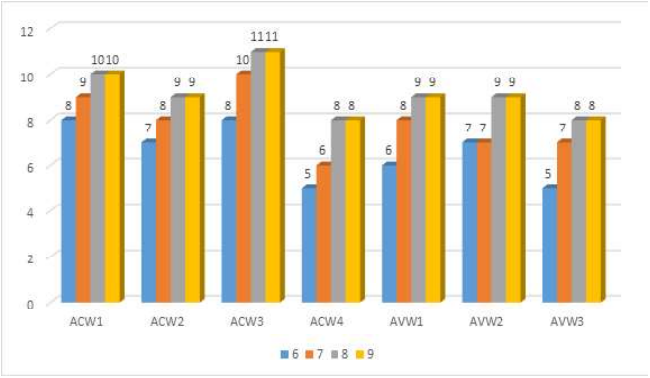


Fig. 2- Graph showing estimation of N_2 production in µg/g soil at different pH.

DISCUSSION

Azotobacter is a nitrogen fixing, phosphate solubilizing, IAA producing, gram negative free living bacteria which colonize in rhizosphere of plants. It is considered as plant growth promoting rhizobacteria.[6] *Azotobacter* has been used as potential nitrogen fertilizer to increase sugar beet yield. *Azotobacter* may be discovered easily from the rhizosphere of cereals.[3] In the present study, isolates of *Azotobacter* were isolated from rhizospheric soil of wheat and maize crops. Among various species of *Azotobacter, A. chroococcum* is the most prevalent species determined in the soil.[5]

Dobereiner (1969)[7] reported that the optimum temperature required for the growth of *Azotobacter* species is 29^{0}C-37^{0}C. In the present study, it was observed that the maximum nitrogen fixation by isolates of *Azotobacter* occurred at a temperature of 34^{0}C. At low and high temperature N_2 production by all isolates decreased. Bakulin *et al.* (2007)[8] and Kisten *et al.* (2006)[9] reported that *Azotobacter* sps. are sensitive to soil properties. In present study, also it was observed that pH directly affected N_2 producing capacity of *Azotobacter* isolates.

CONCLUSION

In the present study, isolates of *Azotobacter* were isolated from rhizospheric soil of wheat and maize field of Madhepura district. Altogether, five villages i.e. Kishanganj, Chousa, Bhelwa, Jeetapur and Shankarpur under Madhepura district were selected from where rhizospheric soil was collected. Altogether, 7 isolates of *Azotobacter* were isolated out of which four belongs to *A. chroococcum* and 3 belongs to *A. vinelandii*. Morphological study, gram staining, biochemical test and sugar fermentation test were performed for all isolates. N_2

producing capacity of all isolates was tested at 3 different temperature (20^0C, 34^0C, 40^0C) and at 4 different pH (6, 7, 8, 9). It was observed that all isolates produced maximum N_2 at a temperature of 34^0C and alkaline pH (pH-8, pH-9). Isolates of *A. chroococcum* were assigned code no. as ACW1, ACW2, ACW3, ACW4 and isolates of *A. vinelandii* as AVW1, AVW2, AVW3. Maximum N_2 was produced by isolates no. ACW3.

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